Multiple-Scales Computations for Proposal 3

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1 Purpose and scope

This note carries out the general calculation proposed for a one-dimensional, translation- and reflection-invariant pattern-forming system near a finite wavenumber Hopf (Turing–Hopf) bifurcation. The calculation is fully explicit in terms of the Fourier matrices of the linear operator and the quadratic and cubic multilinear maps. Numerical values cannot be produced until the specific matrices, kernels, and forcing vectors are supplied.

Two corrections to the handwritten derivation are important from the outset.

1. The modes with spatial wavenumbers $+k_0$ and $-k_0$ have independent complex amplitudes. They must be denoted, for example, by $A(\tau)$ and $B(\tau)$. Setting them equal in advance restricts the calculation to a standing-wave subspace and makes it impossible to decide between standing and traveling waves.

2. If the slow time is $\tau = \epsilon^2 t$, the chain rule is

$$\partial_t = \partial_s + \epsilon^2 \partial_\tau, \qquad s = t, \tag{1}$$

not $\partial_s + \epsilon \partial_\tau$. The latter belongs to the different scaling $\tau = \epsilon t$.

2 Equation, notation, and hypotheses

2.1 Corrected notation

The proposal uses Q both for the quadratic nonlinearity and for a forcing vector. To avoid that collision, write the equation as

$$\begin{aligned} u_t = {} & \mathcal{L}_0 u + p\mathcal{L}_1 u + \mathcal{Q}(u,u) + \mathcal{C}(u,u,u) \\ & + \epsilon^n \left(P \mathrm{e}^{\mathrm{i}\omega t} + \overline{P} \mathrm{e}^{-\mathrm{i}\omega t} \right) + \epsilon^m \left(S \mathrm{e}^{\mathrm{i}kx} + \overline{S} \mathrm{e}^{-\mathrm{i}kx} \right), \end{aligned} \tag{2}$$

where

- $u(x,t) \in \mathbb{R}^d$ and $x \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$;
- p is the bifurcation parameter (called λ in the proposal);
- $\mathcal{Q}$ is a symmetric bilinear map and $\mathcal{C}$ is a symmetric trilinear map, both extended complex linearly during the calculation;
- $P, S \in \mathbb{C}^d$ are the temporal- and spatial-forcing vectors.

Any quadratic or cubic form has a symmetric multilinear polarization, so the symmetry assumptions on $\mathcal{Q}$ and $\mathcal{C}$ lose no generality. If an implementation stores a nonsymmetric bilinear map, every occurrence of $2\mathcal{Q}(a,b)$ below must be replaced by $\mathcal{Q}(a,b) + \mathcal{Q}(b,a)$.

For each integer spatial Fourier mode h, define matrices $M_j(h)$ by

$$\mathcal{L}_j \left(v \mathrm{e}^{\mathrm{i}hx} \right) = M_j(h) v \mathrm{e}^{\mathrm{i}hx}, \qquad j = 0, 1. \tag{3}$$

Meaning and construction of $M_j(h)$. On the 2π-periodic domain, a vector field has the Fourier expansion

$$u(x,t) = \sum_{h\in\mathbb{Z}} \widehat{u}_h(t)\mathrm{e}^{\mathrm{i}hx}, \qquad \widehat{u}_h(t) \in \mathbb{C}^d. \tag{4}$$

Translation invariance prevents the linear operator from mixing different values of h. It can change only the d component amplitudes of a pure mode, so its action on that mode is represented by the $d \times d$ matrix $M_j(h)$. Consequently

$$\mathcal{L}_j u = \sum_{h\in\mathbb{Z}} M_j(h)\widehat{u}_h \mathrm{e}^{\mathrm{i}hx}. \tag{5}$$

The matrices are obtained by applying $\mathcal{L}_j$ to $v\mathrm{e}^{\mathrm{i}hx}$ and using the Fourier replacement rules

$$\partial_x \mapsto \mathrm{i}h, \qquad \partial_{xx} \mapsto -h^2, \qquad W* \mapsto \widehat{W}(h). \tag{6}$$

For example, with $\widehat{W}(h) = \int_0^{2\pi} W(z)\mathrm{e}^{-\mathrm{i}hz}\,dz$,

$$W * \left(a\mathrm{e}^{\mathrm{i}h\,\cdot}\right)(x) = \widehat{W}(h)a\mathrm{e}^{\mathrm{i}hx}. \tag{7}$$

Thus a differential operator $A + D\partial_{xx} + C\partial_x$ has symbol $A - h^2D + \mathrm{i}hC$, while a neural-field convolution contributes the appropriate $\widehat{W}_{ab}(h)$ to each population-to-population matrix entry. If $\mathcal{L}(p) = \mathcal{L}_0 + p\mathcal{L}_1 + O(p^2)$, then similarly $M(h,p) = M_0(h) + pM_1(h) + O(p^2)$, so $M_1(h) = \partial_p M(h,0)$.

Because the original operator is real, complex conjugating (3) gives $M_j(-h) = \overline{M_j(h)}$. If it also commutes with reflection $x \mapsto -x$, then $M_j(-h) = M_j(h)$; for neural fields this usually follows from even coupling kernels. Hence the $+h$ and $-h$ modes have the same linear spectrum, although their amplitudes remain independent until a standing-wave restriction is imposed.

2.2 Finite-wavenumber Hopf hypothesis

Let $k_0 \in \mathbb{Z} \setminus \{0\}$ and $\omega_0 > 0$. Assume

$$M_0(k_0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M_0(k_0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \Psi^*\Phi = 1. \tag{8}$$

Thus Φ is a right eigenvector and Ψ is the adjoint/left eigenvector. The star is the conjugate transpose; a plain transpose is not sufficient after complexifying the problem.

Let $\nu(h,p)$ denote the eigenvalue branch through $\nu(k_0,0) = \mathrm{i}\omega_0$. Standard eigenvalue perturbation gives

$$\partial_p\nu(k_0,0) = \Psi^* M_1(k_0)\Phi. \tag{9}$$

The transversality condition remembered as "$d\lambda/dp > 0$" is more precisely

$$\mathrm{Re}(\Psi^* M_1(k_0)\Phi) > 0. \tag{10}$$

The imaginary part only changes the oscillation frequency to leading order.

We also assume that there are no accidental quadratic resonances: the matrices

$$2\mathrm{i}\omega_0 I - M_0(0), \quad 2\mathrm{i}\omega_0 I - M_0(\pm 2k_0), \quad -M_0(0), \quad -M_0(\pm 2k_0) \tag{11}$$

are invertible. If one is singular, the center space is larger and the two-amplitude normal form derived here is not closed.

3 Scaling and the three order equations

Use the weakly nonlinear Hopf scaling

$$p = \epsilon^2\mu, \qquad s = t, \qquad \tau = \epsilon^2 t, \qquad u = \epsilon u_1 + \epsilon^2 u_2 + \epsilon^3 u_3 + O(\epsilon^4). \tag{12}$$

If one instead writes $p = \epsilon p_1 + \epsilon^2 p_2 + \cdots$, consistency with the slow time $\tau = \epsilon^2 t$ requires $p_1 = 0$ and $p_2 = \mu$. A nonzero p_1 generally creates resonant evolution on the faster scale $\tau = \epsilon t$.

Define

$$\mathcal{D} := \partial_s - \mathcal{L}_0 \tag{13}$$

and let F_j denote the part of the two external forcings whose exponent is ϵ^j:

$$\begin{aligned} F_j = {} & \mathbf{1}_{\{n=j\}} \left(P\mathrm{e}^{\mathrm{i}\omega s} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega s} \right) \\ & + \mathbf{1}_{\{m=j\}} \left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx} \right). \end{aligned} \tag{14}$$

Substitution of (12) into (2), followed by collecting powers of ϵ, gives the complete first three equations:

$$O(\epsilon): \quad \mathcal{D}u_1 = F_1, \tag{15a}$$

$$O(\epsilon^2): \quad \mathcal{D}u_2 = \mathcal{Q}(u_1, u_1) + F_2, \tag{15b}$$

$$O(\epsilon^3): \quad \mathcal{D}u_3 = -\partial_\tau u_1 + \mu\mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2) + \mathcal{C}(u_1, u_1, u_1) + F_3. \tag{15c}$$

These are order equations, sometimes informally called the first, second, and third "moment" equations. Sections 4–8 first derive the unforced normal form. Section 9 then restores every forcing term.

4 The $O(\epsilon)$ solution

Temporarily put $F_1 = 0$. Introduce the fast phases

$$\xi_+ = \omega_0 s + k_0 x, \qquad \xi_- = \omega_0 s - k_0 x, \qquad E_\pm = \mathrm{e}^{\mathrm{i}\xi_\pm}. \tag{16}$$

The four-dimensional real kernel of $\mathcal{D}$ is

$$\ker \mathcal{D} = \mathit{spanop}_{\mathbb{R}} \left\{ \Phi E_+, \Phi E_-, \overline{\Phi}\,\overline{E}_+, \overline{\Phi}\,\overline{E}_- \right\}. \tag{17}$$

Consequently the most general real leading-order solution is

$$u_1 = A(\tau)\Phi E_+ + B(\tau)\Phi E_- + \text{c.c.} \tag{18}$$

or, with the conjugates shown,

$$u_1 = A\Phi E_+ + B\Phi E_- + \overline{A}\,\overline{\Phi}\,\overline{E}_+ + \overline{B}\,\overline{\Phi}\,\overline{E}_-. \tag{19}$$

The amplitudes A and B vary only on the slow time τ. The choice $A = B$ is a standing-wave restriction, while $B = 0$ or $A = 0$ produces one of the two traveling-wave directions.

5 The complete $O(\epsilon^2)$ calculation

Define the two basic quadratic vectors

$$q_{20} := \mathcal{Q}(\Phi, \Phi), \qquad q_{11} := \mathcal{Q}(\Phi, \overline{\Phi}). \tag{20}$$

For a real symmetric quadratic form, q_{11} is real. Direct multiplication of all four terms in (19) gives

$$\begin{aligned}\mathcal{Q}(u_1, u_1) = q_{20}\left(A^2E_+^2 + B^2E_-^2 + 2ABE_+E_-\right) + \text{c.c.}\\ + 2q_{11}\left(|A|^2 + |B|^2 + A\overline{B}E_+\overline{E}_- + \overline{A}B\overline{E}_+E_-\right).\end{aligned} \tag{21}$$

The phase products are

$$\begin{aligned}E_+^2 = \mathrm{e}^{2\mathrm{i}(\omega_0 s + k_0 x)}, \qquad E_-^2 = \mathrm{e}^{2\mathrm{i}(\omega_0 s - k_0 x)},\\ E_+E_- = \mathrm{e}^{2\mathrm{i}\omega_0 s}, \qquad E_+\overline{E}_- = \mathrm{e}^{2\mathrm{i}k_0 x}, \qquad \overline{E}_+E_- = \mathrm{e}^{-2\mathrm{i}k_0 x}.\end{aligned} \tag{22}$$

Thus the quadratic nonlinearity generates precisely the harmonics

$$(\Omega, h) = (\pm 2\omega_0, \pm 2k_0), \quad (\pm 2\omega_0, 0), \quad (0, \pm 2k_0), \quad (0, 0). \tag{23}$$

None is critical under the nonresonance hypothesis.

For later use define the resolvent

$$R(\Omega, h) := (\mathrm{i}\Omega I - M_0(h))^{-1}. \tag{24}$$

Define the second-order response vectors

$$h_{20}^+ = R(2\omega_0, 2k_0)q_{20}, \qquad h_{20}^- = R(2\omega_0, -2k_0)q_{20}, \tag{25a}$$

$$h_{20}^0 = 2R(2\omega_0, 0)q_{20}, \qquad h_{11}^0 = 2R(0, 0)q_{11}, \tag{25b}$$

$$h_{11}^+ = 2R(0, 2k_0)q_{11}, \qquad h_{11}^- = 2R(0, -2k_0)q_{11}. \tag{25c}$$

The factors of two in (25) come directly from (21). Since

$$\mathcal{D}\left(v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (\mathrm{i}\Omega I - M_0(h))\, v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}, \tag{26}$$

a particular solution of (15b), with no kernel component added, is

$$\begin{aligned}u_2 = A^2h_{20}^+E_+^2 + B^2h_{20}^-E_-^2 + ABh_{20}^0E_+E_- + \text{c.c.}\\ + (|A|^2 + |B|^2)h_{11}^0 + A\overline{B}h_{11}^+E_+\overline{E}_- + \overline{A}Bh_{11}^-\overline{E}_+E_-.\end{aligned} \tag{27}$$

Reality and reflection symmetry imply the needed conjugacy relations among $h_{20}^\pm$ and $h_{11}^\pm$. A homogeneous kernel contribution to u_2 can be absorbed into a near-identity redefinition of A and B, so the choice (27) is the standard normalization.

6 Fredholm alternative and the adjoint double integral

Let

$$T_0 = \frac{2\pi}{\omega_0} \tag{28}$$

and use the normalized space–time inner product

$$\langle v, w \rangle := \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} v(s,x)^* w(s,x)\, dx\, ds. \tag{29}$$

Periodic integration by parts shows that

$$\mathcal{D}^* = -\partial_s - \mathcal{L}_0^*. \tag{30}$$

Indeed, the adjoint critical functions

$$V_+ = \Psi E_+, \qquad V_- = \Psi E_- \tag{31}$$

satisfy $\mathcal{D}^* V_\pm = 0$ because

$$(-\mathrm{i}\omega_0 I - M_0(k_0)^*)\, \Psi = 0. \tag{32}$$

If one defines the leading operator with the opposite sign, $\widetilde{\mathcal{D}} = \mathcal{L}_0 - \partial_s$, its adjoint also changes by an overall sign. This explains handwritten variants resembling $-\partial_s + M_0(k_0)^T$; all variants give the same solvability condition when used consistently.

The Fredholm alternative says that

$$\mathcal{D} w = f \tag{33}$$

has a T_0-periodic-in-s, 2π-periodic-in-x solution only if f is orthogonal to every element of $\ker \mathcal{D}^*$. At third order this means

$$0 = \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} \Psi^* \mathrm{e}^{-\mathrm{i}(\omega_0 s + k_0 x)} F^{(3)}(s,x)\, dx\, ds, \tag{34a}$$

$$0 = \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} \Psi^* \mathrm{e}^{-\mathrm{i}(\omega_0 s - k_0 x)} F^{(3)}(s,x)\, dx\, ds, \tag{34b}$$

where, without forcing for the moment,

$$F^{(3)} = -\partial_\tau u_1 + \mu \mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2) + \mathcal{C}(u_1, u_1, u_1). \tag{35}$$

The two integrals simply extract the coefficients of E_+ and E_- and then left-multiply those vector coefficients by Ψ^*.

7 The complete resonant $O(\epsilon^3)$ terms

Only terms proportional to E_+ enter (34a); terms proportional to E_- enter (34b). Orthogonality of all other Fourier harmonics makes them integrate to zero.

7.1 Slow derivative and parameter terms

The E_+ coefficient is

$$[-\partial_\tau u_1]_{E_+} = -A_\tau \Phi, \qquad [\mu \mathcal{L}_1 u_1]_{E_+} = \mu A M_1(k_0) \Phi. \tag{36}$$

The corresponding E_- terms are obtained by replacing A by B.

7.2 Cubic term

To obtain E_+ from three first-order phases, the allowed products are

$$E_+E_+\overline{E}_+ = E_+, \qquad E_+E_-\overline{E}_- = E_+. \tag{37}$$

There are three permutations of the first product and six permutations of the second. Hence

$$\begin{aligned}[\mathcal{C}(u_1,u_1,u_1)]_{E_+} &= 3A|A|^2\mathcal{C}(\Phi,\Phi,\overline{\Phi}) \\ &\quad + 6A|B|^2\mathcal{C}(\Phi,\Phi,\overline{\Phi}).\end{aligned} \tag{38}$$

7.3 Quadratic feedback through u_2

The complete list of phase pairings from u_1u_2 that produce E_+ is

$$\begin{aligned} E_+\cdot 1 &= E_+, & \overline{E}_+\cdot E_+^2 &= E_+, \\ \overline{E}_-\cdot(E_+E_-) &= E_+, & E_-\cdot(E_+\overline{E}_-) &= E_+. \end{aligned}$$

Using (27), these four pairings give

$$\begin{aligned}[2\mathcal{Q}(u_1,u_2)]_{E_+} &= 2A(|A|^2+|B|^2)\mathcal{Q}(\Phi,h_{11}^0) \\ &\quad + 2A|A|^2\mathcal{Q}(\overline{\Phi},h_{20}^+) \\ &\quad + 2A|B|^2\mathcal{Q}(\overline{\Phi},h_{20}^0) \\ &\quad + 2A|B|^2\mathcal{Q}(\Phi,h_{11}^+).\end{aligned} \tag{39}$$

It is useful to collect the self- and cross-interaction vectors:

$$G_{\mathrm{s}} = 2\mathcal{Q}(\Phi,h_{11}^0) + 2\mathcal{Q}(\overline{\Phi},h_{20}^+) + 3\mathcal{C}(\Phi,\Phi,\overline{\Phi}), \tag{40a}$$

$$G_{\mathrm{c}} = 2\mathcal{Q}(\Phi,h_{11}^0) + 2\mathcal{Q}(\overline{\Phi},h_{20}^0) + 2\mathcal{Q}(\Phi,h_{11}^+) + 6\mathcal{C}(\Phi,\Phi,\overline{\Phi}). \tag{40b}$$

Therefore the full E_+ vector coefficient in (35) is

$$-A_\tau\Phi + \mu A M_1(k_0)\Phi + A|A|^2G_{\mathrm{s}} + A|B|^2G_{\mathrm{c}}. \tag{41}$$

Reflection symmetry produces the E_- coefficient

$$-B_\tau\Phi + \mu B M_1(k_0)\Phi + B|B|^2G_{\mathrm{s}} + B|A|^2G_{\mathrm{c}}. \tag{42}$$

8 Unforced amplitude equations

Apply Ψ^* to (41) and (42), use $\Psi^*\Phi = 1$, and define

$$a := \mu\Psi^*M_1(k_0)\Phi, \qquad b := \Psi^*G_{\mathrm{s}}, \qquad c := \Psi^*G_{\mathrm{c}}. \tag{43}$$

The Turing–Hopf normal form is

$$A_\tau = A\left(a + b|A|^2 + c|B|^2\right), \tag{44a}$$

$$B_\tau = B\left(a + b|B|^2 + c|A|^2\right). \tag{44b}$$

The coefficients are generally complex. Their imaginary parts correct the temporal phases/frequencies; their real parts determine amplitudes and stability.

Writing $A = r_+ e^{i\theta_+}$ and $B = r_- e^{i\theta_-}$ yields

$$(r_+)_\tau = r_+ \left(a_r + b_r r_+^2 + c_r r_-^2\right), \tag{45a}$$
$$(r_-)_\tau = r_- \left(a_r + b_r r_-^2 + c_r r_+^2\right), \tag{45b}$$

where $a_r = \mathrm{Re}(a)$, $b_r = \mathrm{Re}(b)$, and $c_r = \mathrm{Re}(c)$.

For $a_r > 0$ just beyond onset:

- A traveling wave has $(r_+, r_-) = (\rho, 0)$ or $(0, \rho)$ with

$$\rho^2 = -\frac{a_r}{b_r}. \tag{46}$$

  It exists supercritically if $b_r < 0$ and is stable to the opposite traveling mode if $c_r - b_r < 0$.

- A standing wave has $r_+ = r_- = \rho$ with

$$\rho^2 = -\frac{a_r}{b_r + c_r}. \tag{47}$$

  It exists supercritically if $b_r + c_r < 0$ and is stable to amplitude imbalance if $c_r - b_r > 0$.

To leading order, the physical patterns are

$$u_{\mathrm{TW}} \simeq 2\epsilon\rho\, \mathrm{Re}\Big(\Phi e^{i(\omega_0 t \pm k_0 x + \theta)}\Big), \tag{48}$$
$$u_{\mathrm{SW}} \simeq 4\epsilon\rho \cos(k_0 x + \vartheta_x)\, \mathrm{Re}\Big(\Phi e^{i(\omega_0 t + \vartheta_t)}\Big). \tag{49}$$

9 External forcing and resonance selection

9.1 A bookkeeping rule that retains every term

Associate a fast frequency–wavenumber pair with each first-order factor:

factor	A	B	$\overline{A}$	$\overline{B}$	P	S
(Ω, h)	(ω_0, k_0)	$(\omega_0, -k_0)$	$(-\omega_0, -k_0)$	$(-\omega_0, k_0)$	$(\omega, 0)$	$(0, k)$

(50)

Conjugate forcing factors have pairs $(-\omega, 0)$ and $(0, -k)$. Multiplication adds the pairs. A term is resonant in the A equation exactly when its sum is (ω_0, k_0), and it is resonant in the B equation exactly when its sum is $(\omega_0, -k_0)$.

There is also an amplitude-order rule. Each critical factor costs one power of ϵ, each P or $\overline{P}$ costs n powers, and each S or $\overline{S}$ costs m powers. A term competes with the cubic normal form when the total cost is three.

For computer algebra or a model-specific calculation, write $\widehat{u}_j(\Omega, h)$ for the coefficient of $e^{i(\Omega s + hx)}$ in u_j. For every noncritical harmonic,

$$\widehat{u}_2(q) = R(q)\Big[\sum_{q_1+q_2=q} \mathcal{Q}(\widehat{u}_1(q_1), \widehat{u}_1(q_2)) + \widehat{F}_2(q)\Big], \tag{51}$$

where $q = (\Omega, h)$ and $R(q) = R(\Omega, h)$. At a critical harmonic $q_c = (\omega_0, \pm k_0)$, the complete third-order vector is

$$\begin{aligned}\widehat{F}^{(3)}(q_c) = & -\partial_\tau \widehat{u}_1(q_c) + \mu M_1(\pm k_0)\widehat{u}_1(q_c) + \widehat{F}_3(q_c) \\ & + \sum_{q_1+q_2=q_c} [\mathcal{Q}(\widehat{u}_1(q_1), \widehat{u}_2(q_2)) + \mathcal{Q}(\widehat{u}_2(q_1), \widehat{u}_1(q_2))] \\ & + \sum_{q_1+q_2+q_3=q_c} \mathcal{C}(\widehat{u}_1(q_1), \widehat{u}_1(q_2), \widehat{u}_1(q_3)). \end{aligned} \tag{52}$$

The Fredholm condition is simply $\Psi^* \widehat{F}^{(3)}(q_c) = 0$. Equations (51)–(52) are an exhaustive recipe: they include the many nonresonant forcing products, forcing-induced mean fields, and coefficient shifts that are easy to miss by hand.

9.2 Why neither separated forcing is directly resonant

The temporal forcing has pair $(\omega, 0)$. Even when $\omega = \omega_0$, its spatial projection onto $\mathrm{e}^{\pm \mathrm{i} k_0 x}$ vanishes because $k_0 \neq 0$. The spatial forcing has pair $(0, k)$. Even when $k = k_0$, its temporal projection onto $\mathrm{e}^{\mathrm{i}\omega_0 s}$ vanishes because $\omega_0 \neq 0$. Therefore neither separated forcing appears as a direct additive term in (44). Nonlinearity must mix harmonics.

For a noncritical external harmonic define its linear response vectors

$$p_\omega := R(\omega, 0)P, \qquad s_k^+ := R(0, k)S, \qquad s_k^- := R(0, -k)\overline{S}. \tag{53}$$

9.3 Temporal 2:1 parametric resonance

Let the temporal forcing be $O(\epsilon^2)$, so $n = 2$, and take

$$\omega = 2\omega_0 + \epsilon^2 \delta_t. \tag{54}$$

Its second-order response is $p_2 \mathrm{e}^{2\mathrm{i}\omega_0 s} \mathrm{e}^{\mathrm{i}\delta_t \tau}$ with

$$p_2 = R(2\omega_0, 0)P. \tag{55}$$

The products

$$(2\omega_0, 0) + (-\omega_0, k_0) = (\omega_0, k_0), \tag{56}$$

and

$$(2\omega_0, 0) + (-\omega_0, -k_0) = (\omega_0, -k_0) \tag{57}$$

couple conjugate counter-propagating modes. The forced normal form acquires

$$A_\tau = A(a + b|A|^2 + c|B|^2) + \eta_t \mathrm{e}^{\mathrm{i}\delta_t \tau} \overline{B}, \tag{58a}$$

$$B_\tau = B(a + b|B|^2 + c|A|^2) + \eta_t \mathrm{e}^{\mathrm{i}\delta_t \tau} \overline{A}, \tag{58b}$$

where, for symmetric $\mathcal{Q}$,

$$\eta_t = 2\Psi^* \mathcal{Q}(\overline{\Phi}, p_2). \tag{59}$$

Thus temporal forcing at twice the Hopf frequency parametrically locks pairs of counter-propagating waves. The exponent $n = 2$ is required because the forcing response multiplies one $O(\epsilon)$ critical mode and must enter at $O(\epsilon^3)$.

9.4 Spatial 2:1 parametric resonance

Let the spatial forcing be $O(\epsilon^2)$, so $m = 2$, and choose $k = 2k_0$. Put

$$s_2^+ = R(0, 2k_0)S, \qquad s_2^- = R(0, -2k_0)\overline{S}. \tag{60}$$

Because

$$(0, 2k_0) + (\omega_0, -k_0) = (\omega_0, k_0), \tag{61}$$

and the conjugate relation holds for the other direction, the normal form is

$$A_\tau = A(a + b|A|^2 + c|B|^2) + \eta_x^+ B, \tag{62a}$$
$$B_\tau = B(a + b|B|^2 + c|A|^2) + \eta_x^- A, \tag{62b}$$

with

$$\eta_x^+ = 2\Psi^* \mathcal{Q}(\Phi, s_2^+), \qquad \eta_x^- = 2\Psi^* \mathcal{Q}(\Phi, s_2^-). \tag{63}$$

For a real cosine forcing in a reflection-symmetric system, these coefficients obey the corresponding conjugacy relation. This term linearly scatters a right-traveling mode into a left-traveling mode and tends to pin standing waves to the imposed spatial pattern.

9.5 Mixed additive spatio-temporal resonance

The product of the temporal and spatial responses has pairs $(\omega, \pm k)$. It is additive at the critical harmonic when

$$\omega = \omega_0 + \epsilon^2 \delta, \qquad k = k_0. \tag{64}$$

To enter at the same $O(\epsilon^3)$ order as the cubic normal form, its power must satisfy

$$n + m = 3. \tag{65}$$

For positive natural numbers this gives $(n, m) = (1, 2)$ or $(2, 1)$. Define

$$p_0 = R(\omega_0, 0)P, \qquad s_0^+ = R(0, k_0)S, \qquad s_0^- = R(0, -k_0)\overline{S}. \tag{66}$$

The additive terms are

$$f_+ = 2\Psi^* \mathcal{Q}(p_0, s_0^+), \qquad f_- = 2\Psi^* \mathcal{Q}(p_0, s_0^-). \tag{67}$$

The normal form therefore contains

$$A_\tau = A(a + b|A|^2 + c|B|^2) + f_+ \mathrm{e}^{\mathrm{i}\delta\tau} + \text{other allowed forcing terms}, \tag{68a}$$
$$B_\tau = B(a + b|B|^2 + c|A|^2) + f_- \mathrm{e}^{\mathrm{i}\delta\tau} + \text{other allowed forcing terms}. \tag{68b}$$

Here "other allowed forcing terms" matters when one forcing is $O(\epsilon)$: two copies of that response together with one critical mode also cost three powers of ϵ and can shift the linear coefficient. Those shifts are generated without omission by (51)–(52).

9.6 Summary of the leading resonance balances

Mechanism	Exponent balance	Frequency/wavenumber condition	Amplitude-equation effect
Unforced saturation	$1+1+1=3$	automatic	$A\lvert A\rvert^2$, $A\lvert B\rvert^2$ and reflected terms
Parameter variation	$2+1=3$	critical mode	aA, aB
Temporal parametric forcing	$n+1=3$, hence $n=2$	$\omega = 2\omega_0 + O(\epsilon^2)$	$\overline{B}$ in the A equation and $\overline{A}$ in the B equation
Spatial parametric forcing	$m+1=3$, hence $m=2$	$k = 2k_0$	B in the A equation and A in the B equation
Mixed additive forcing	$n+m=3$	$\omega = \omega_0 + O(\epsilon^2)$, $k = k_0$	constants $f_{\pm}\mathrm{e}^{\mathrm{i}\delta\tau}$
Forcing-induced linear shift	$2n+1=3$ or $2m+1=3$	forcing and its conjugate have zero net pair	$\lvert P\rvert^2 A$ for $n=1$, or $\lvert S\rvert^2 A$ for $m=1$, and reflected terms

More complicated resonances require no new principle. Form every allowed quadratic or cubic monomial, add its frequency–wavenumber pairs from (50), check whether the result is $(\omega_0, \pm k_0)$, and check whether its ϵ cost is three. If a resonant term appears already at order two, the $\tau = \epsilon^2 t$ scaling is inconsistent; the forcing must be weakened or a faster slow time must be introduced.

10 Specialization to the excitatory–inhibitory neural field

For the neural field in the proposal, write $U = (u, v)^T$ and linearize about a spatially uniform equilibrium. Let $f_e^{(r)}$ and $f_i^{(r)}$ denote the rth derivatives of the firing-rate function evaluated at the excitatory and inhibitory equilibrium inputs. With Fourier coefficients $\widehat{W}_{ab}(h)$, the linear Fourier matrix is

$$M_0(h) = \begin{pmatrix} \dfrac{-1 + f_e'\widehat{W}_{ee}(h)}{\tau_e} & \dfrac{-f_e'\widehat{W}_{ie}(h)}{\tau_e} \\ \dfrac{f_i'\widehat{W}_{ei}(h)}{\tau_i} & \dfrac{-1 - f_i'\widehat{W}_{ii}(h)}{\tau_i} \end{pmatrix}. \tag{69}$$

For fields $U = (u, v)^T$ and $V = (\widetilde{u}, \widetilde{v})^T$, define

$$\ell_e(U) = W_{ee} * u - W_{ie} * v, \qquad \ell_i(U) = W_{ei} * u - W_{ii} * v. \tag{70}$$

Then the polarized Taylor terms are

$$\mathcal{Q}(U, V) = \begin{pmatrix} \dfrac{f_e''}{2\tau_e}\ell_e(U)\ell_e(V) \\ \dfrac{f_i''}{2\tau_i}\ell_i(U)\ell_i(V) \end{pmatrix}, \tag{71}$$

and

$$\mathcal{C}(U, V, W) = \begin{pmatrix} \dfrac{f_e'''}{6\tau_e}\ell_e(U)\ell_e(V)\ell_e(W) \\ \dfrac{f_i'''}{6\tau_i}\ell_i(U)\ell_i(V)\ell_i(W) \end{pmatrix}. \tag{72}$$

Equations (69)–(72), together with the choice of bifurcation parameter that determines $M_1(h)$, provide every object needed in (25), (40), and (43).

11 Model-specific computation checklist

For a concrete parameter set, the calculation can be completed in the following order.

1. Compute $M_0(h)$ and scan integer h for a simple pair $\pm i\omega_0$ crossing at $h = \pm k_0$.
2. Compute Φ and Ψ from (8), then normalize $\Psi^* \Phi = 1$.
3. Verify the transversality condition (10) and the nonresonance/invertibility conditions (11).
4. Evaluate q_{20} and q_{11} from (20).
5. Solve the six small linear systems in (25); explicit matrix inversion is unnecessary and numerically less stable.
6. Form G_{s} and G_{c} from (40), then project with Ψ^* to obtain a, b, c.
7. Compare $\mathrm{Re}(b)$ and $\mathrm{Re}(c)$ using the criteria in Section 8.
8. Add the desired forcing harmonics using (51)–(52). The resonance table determines which choices of n, m, ω, k affect the leading amplitude equations.

12 What cannot be made numerical yet

The derivation above is complete at the abstract level, but the following data are required for numerical coefficients:

- the exact entries of $M_0(h)$ and $M_1(h)$, or equivalently all neural kernels, time constants, equilibrium values, and the parameter being varied;
- the components and phases of the forcing vectors P and S;
- the intended temporal and spatial resonance, including any detuning;
- the convention for Fourier coefficients of the convolution kernels.

Once these are supplied, all remaining work consists of finite-dimensional linear solves and inner products listed in the checklist.

References

[1] R. Curtu and B. Ermentrout, "Pattern Formation in a Network of Excitatory and Inhibitory Cells with Adaptation," *SIAM Journal on Applied Dynamical Systems*, 3(3), 191–231, 2004. doi:10.1137/030600503.